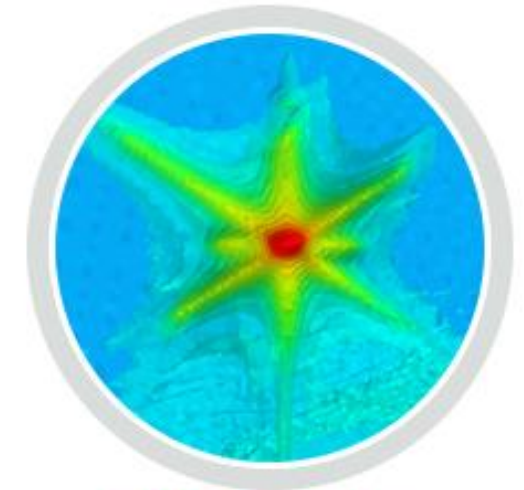
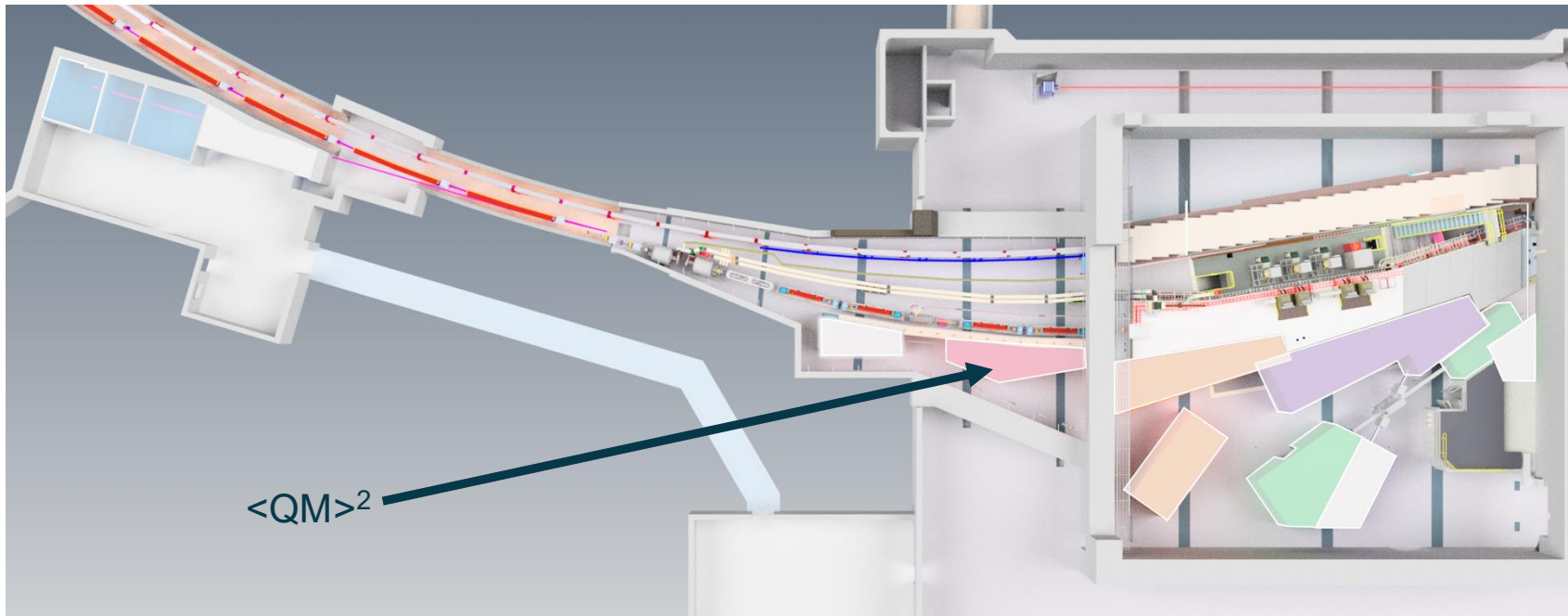


Q-Mapping for Quantum Materials $\langle QM \rangle^2$ Beamline; ID4B



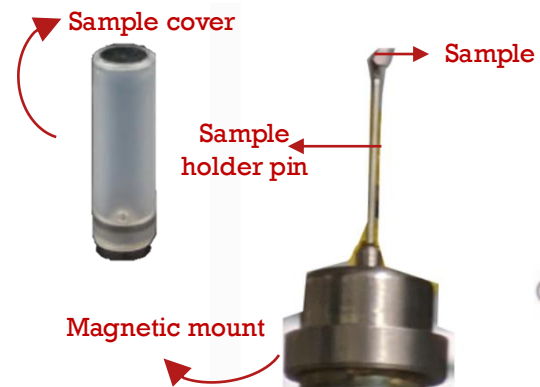
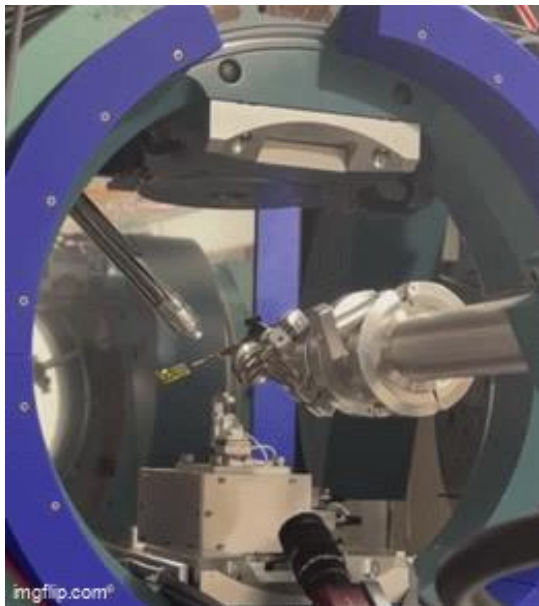
QM2 Beamline

High-throughput characterization of quantum materials.



Single crystal mount

3



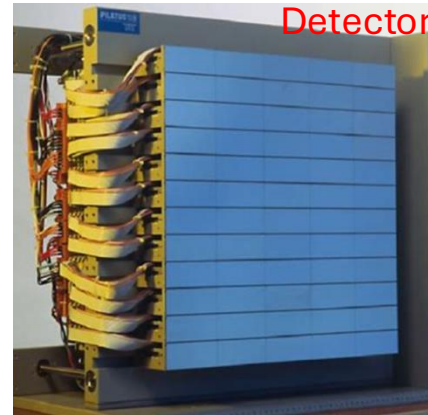
MiTeGen



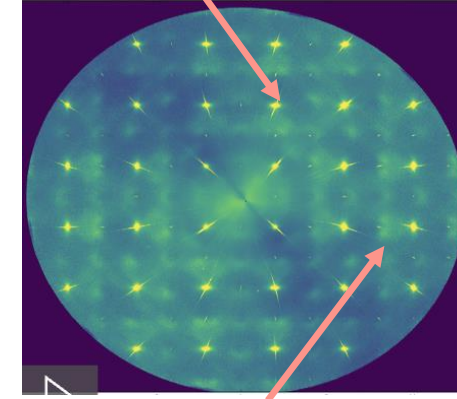
Dual Thickness MicroMounts - 100
SKU: M2-L18SP-100



<https://suchismitasarker.github.io/CHESS-ID4B-QM2/sample/>

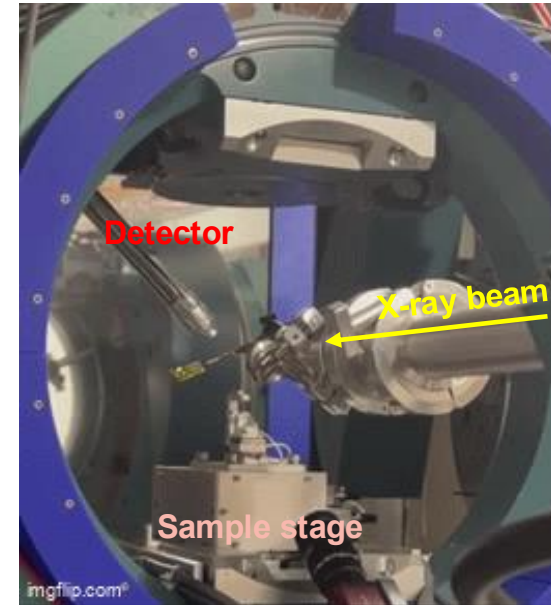


Bragg peak
(Average structure)

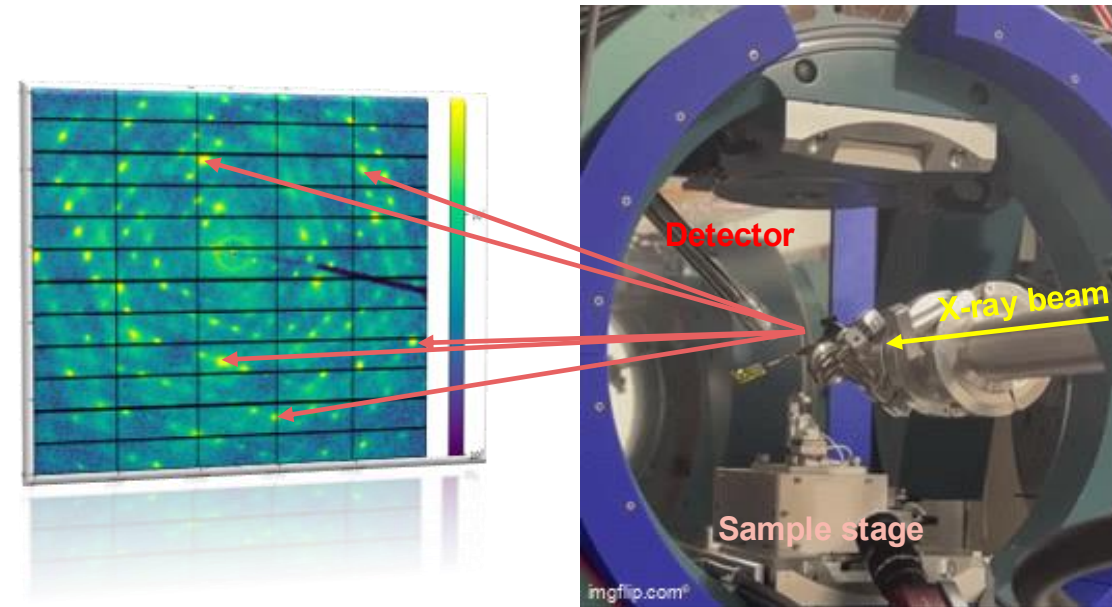


Diffuse Scattering
(Deviation from average structure)

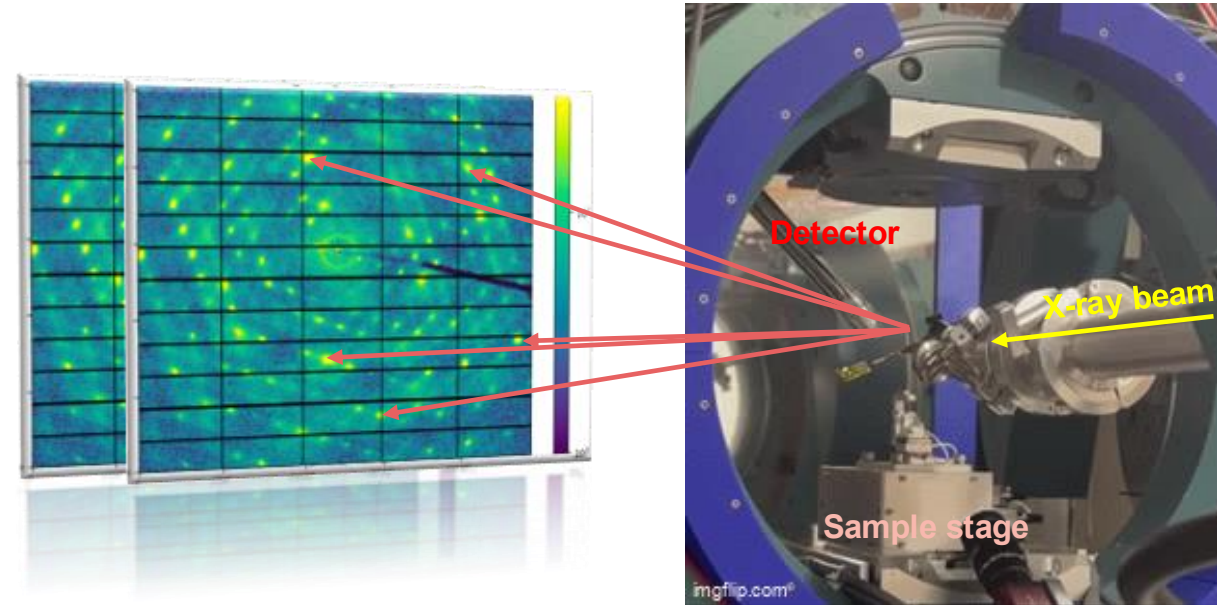
How do we collect the data at 4B?



How do we collect the data at 4B?

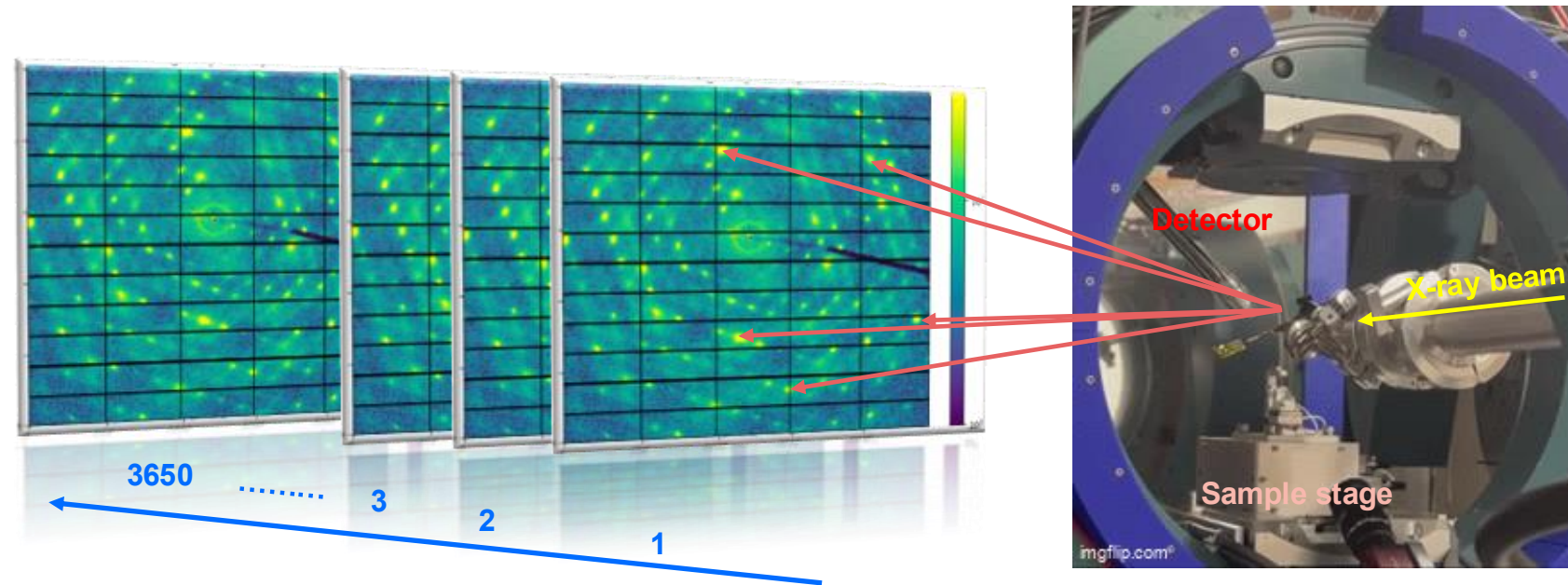


How do we collect the data at 4B?



How do we collect the data at 4B?

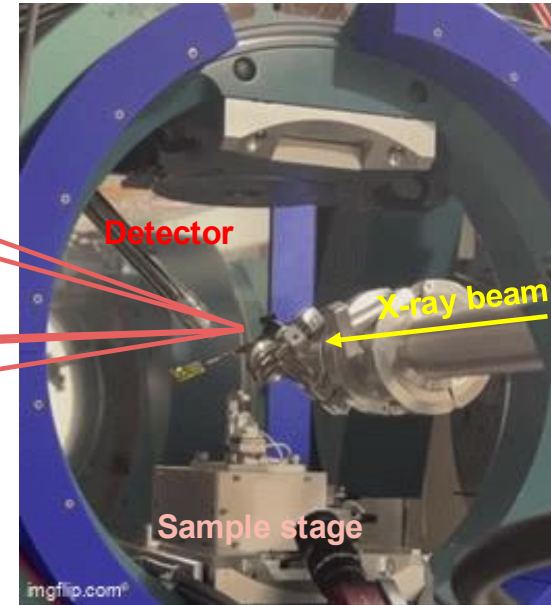
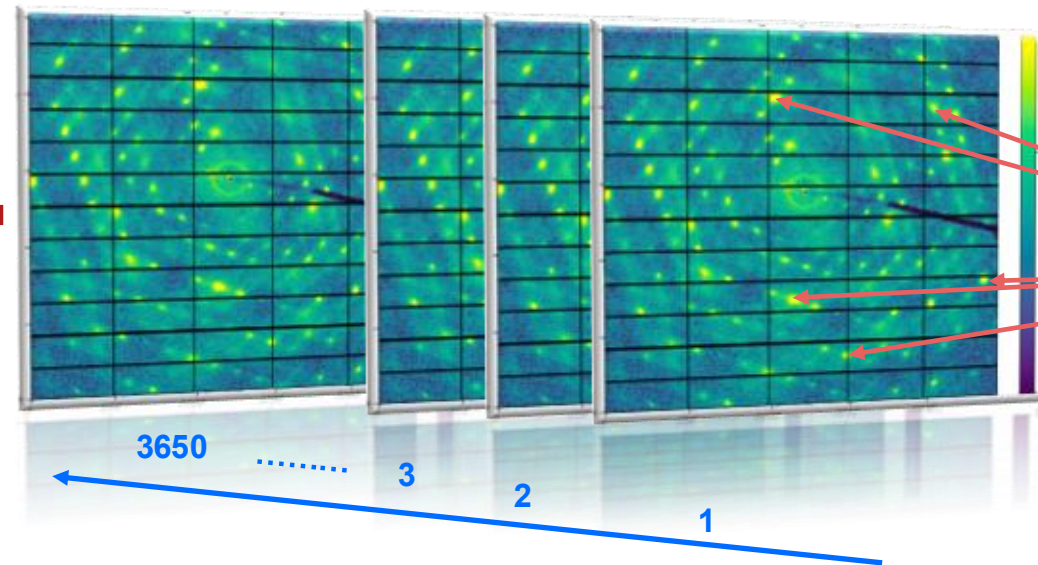
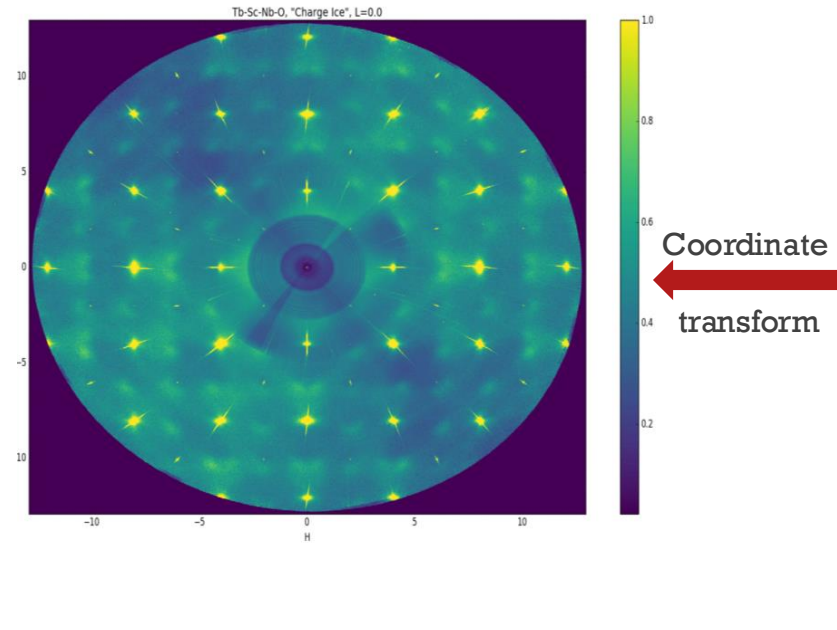
Data collection: 30 mins



How do we collect the data at 4B?

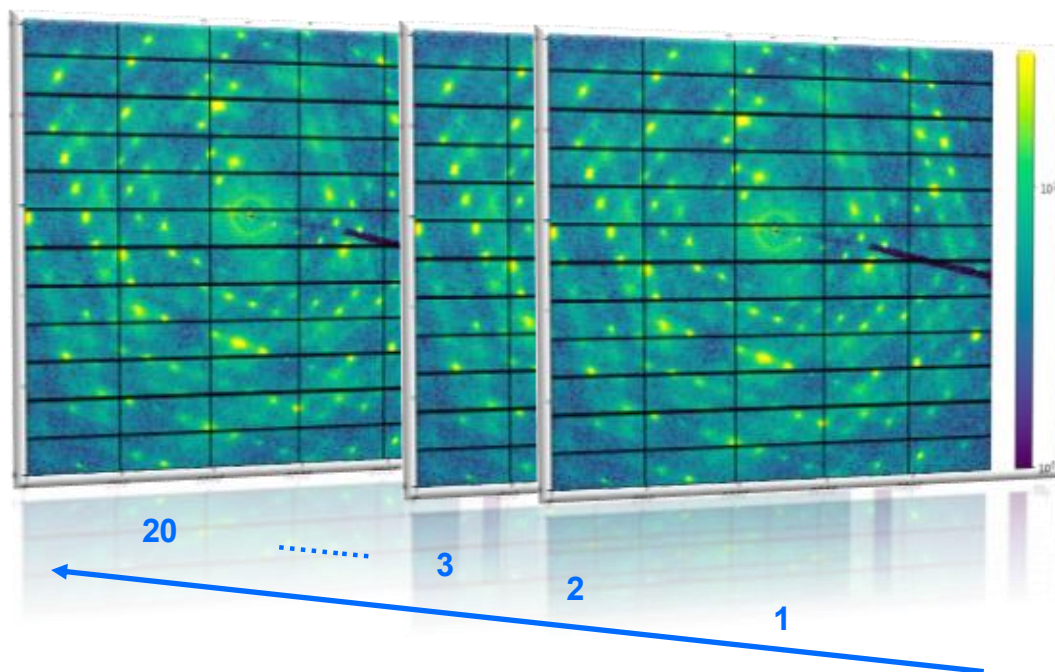
Data processing : 4 hours

Data collection: 20 mins

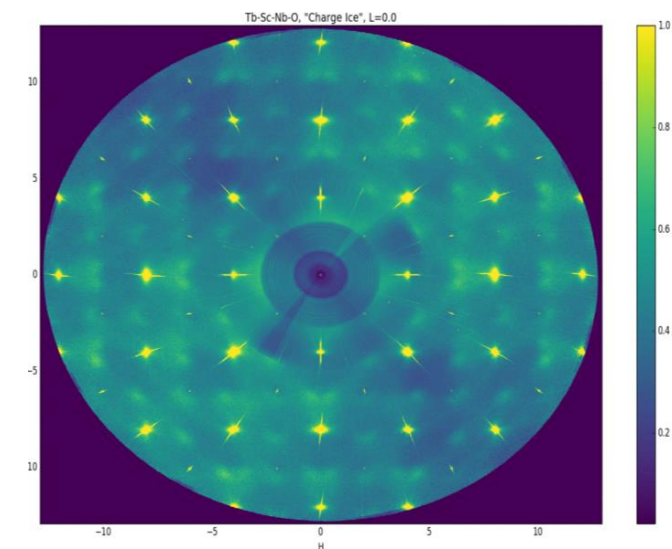


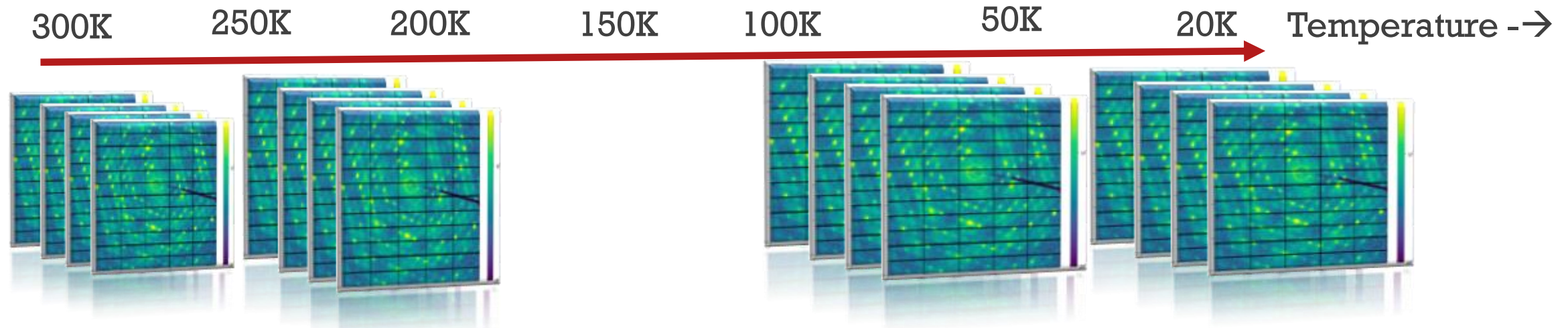
Total frame : 10950 images

Raw data



Processed data





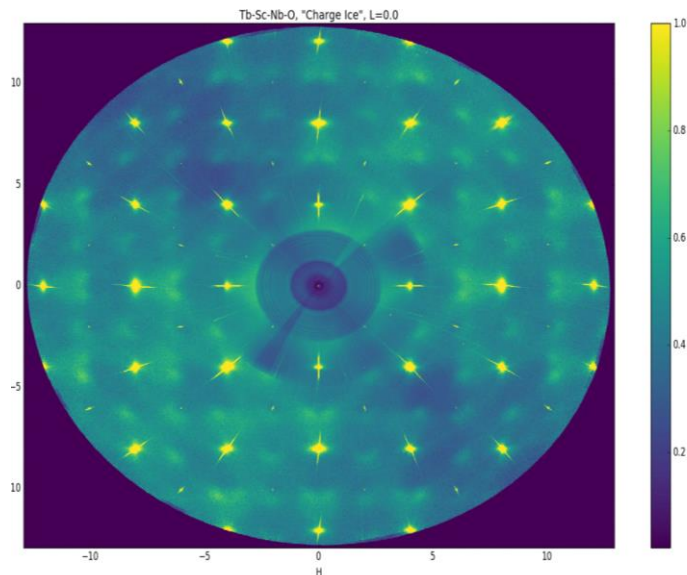
It answers:

“What is the best setting to achieve optimal performance?”

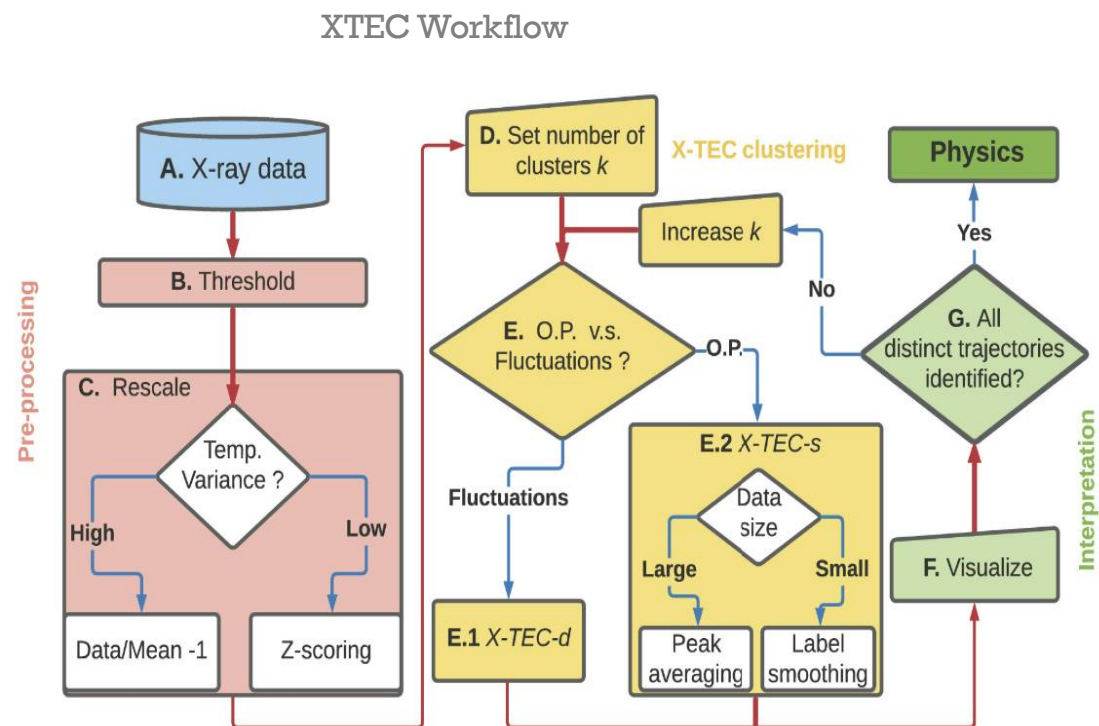
- Goal: find temperature where to collect the data?
- Ask: “Which temperature is most promising AND uncertain?”

Bayesian Optimization is a method to **find the best input that maximizes or minimizes a function** when evaluations are expensive.

XTEC

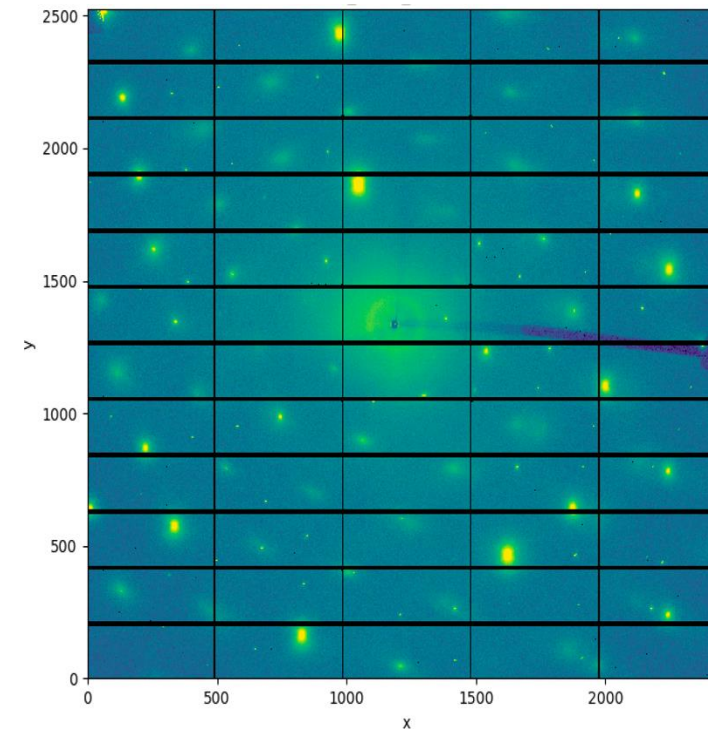
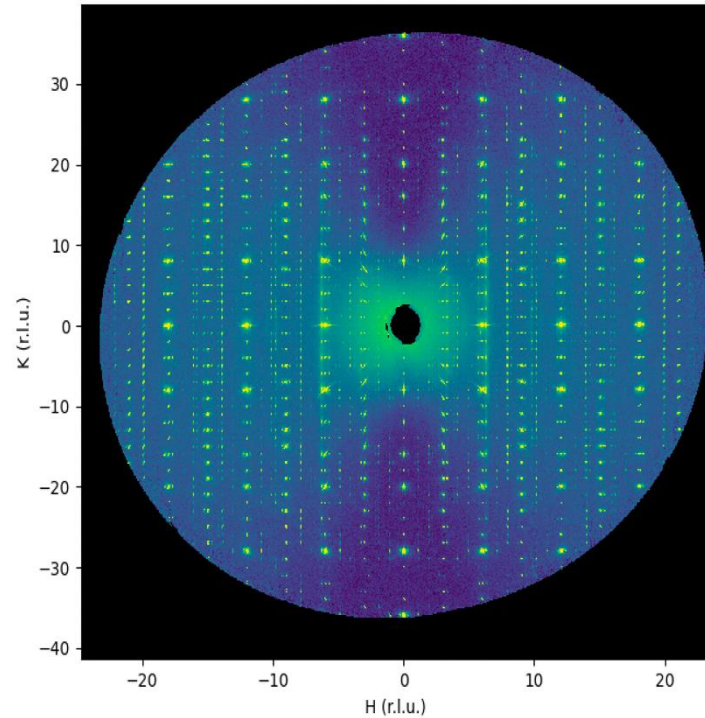


X-TEC has been successfully used for unsupervised learning of phase transitions and identification of key features in $I(H,K,L)$ reciprocal space volumes. However, data indexing and transformation prior to study by X-TEC can require hours of pre-processing. The utility of “immediate” X-TEC pre-analysis on untransformed raw data has not yet been explored. It could be very powerful.



Use different clustering algorithm both processed and raw data

Use Clustering algorithm to learn temperature trajectories of untransformed datasets, identifying phase transitions in real time during data collection, and using clustering to inform data processing steps (peak identification, symmetry determination, masking, etc)



Next week: Task

Tasks: Learn the fundamentals of key algorithms

Aditya: Bayesian Optimization & Active Learning

- Concepts of algorithm
- Feature extraction and feature selections
- Workflow
- How the real time measurement can be done with BO and active learning

Olivia & Max: Clustering

- Clustering concepts
 - What is clustering, types (example: K-means, DBSCAN, GMM, Hierarchical,)
 - when to use
-
- Configure computational environment
 - Gain access to CHES clusters / HPC systems
 - Install packages after reading materials